

①  $X \sim \text{Bernoulli}(p)$

$$X=1, p$$

$$X=0, 1-p$$

$$\mathbb{E}[X] = 1 \times p + 0 \times (1-p) = p$$

$$\begin{aligned} \mathbb{V}[X] &= (1-p)^2 \times p + (0-p)^2 \times (1-p) \\ &= p(1-p)^2 + p^2(1-p) \\ &= p(1-p)(1-p+p) = p(1-p) \end{aligned}$$

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$$P(X=x) = \binom{n}{x} \times p^x \times (1-p)^{n-x}$$

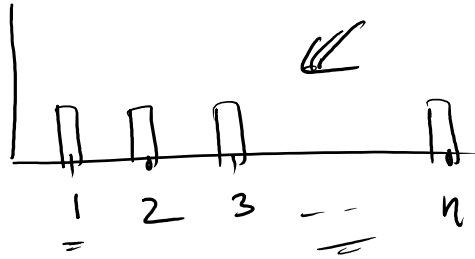
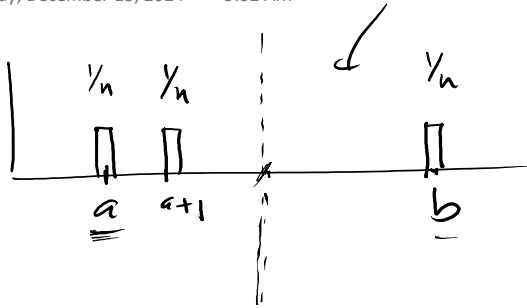
$$Y = X_1 + X_2 + \dots + X_n, \quad X_i \sim \text{Bernoulli}(p)$$

$$\mathbb{E}[Y] = \mathbb{E}[X_1 + X_2 + \dots + X_n]$$

$$= \mathbb{E}(X_1) + \mathbb{E}(X_2) + \dots + \mathbb{E}(X_n) = np$$

$$\mathbb{V}[Y] = \mathbb{V}[X_1 + X_2 + \dots + X_n]$$

$$= \mathbb{V}(X_1) + \mathbb{V}(X_2) + \dots + \mathbb{V}(X_n) = n p(1-p)$$



$$\eta = b - a + 1$$

$$E[X] = \frac{a+b}{2}$$

$$E[X] = a \times \frac{1}{\eta} + (a+1) \frac{1}{\eta} + \dots + b \cdot \left(\frac{1}{\eta}\right)$$

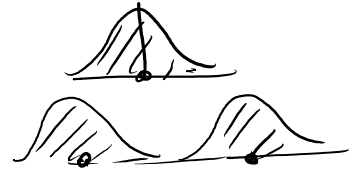
$$V(X) = \sum \left( x - \frac{1+\eta}{2} \right)^2 \times \left( \frac{1}{\eta} \right)$$

$$= \frac{1}{\eta} \left[ \sum \left( x^2 + \left( \frac{\eta+1}{2} \right)^2 - x(\eta+1) \right) \right]$$

$$= \frac{1}{\eta} \left[ \frac{n(n+1)(2n+1)}{6} + n \cdot \frac{(\eta+1)^2}{4} - (n+1) \cdot \frac{n(n+1)}{2} \right]$$

$$= \frac{1}{\eta} \left[ \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{4} \right]$$

$$= \frac{(\eta+1)}{12} \left( 4\eta+2 - 3\eta-3 \right) = \frac{(\eta+1)(\eta-1)}{12} = \frac{\eta^2-1}{12}$$

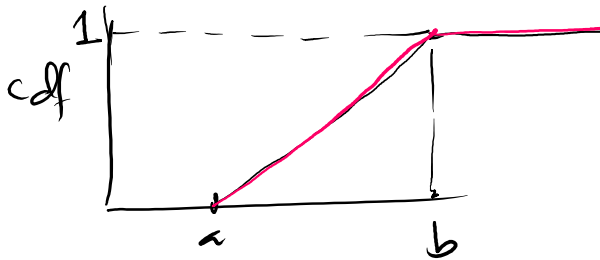
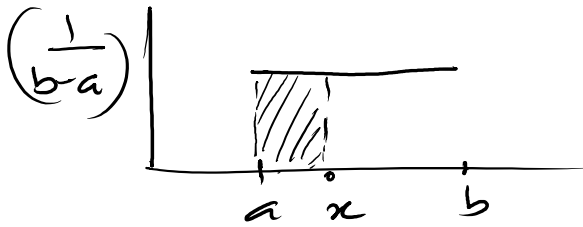


Q.3.  $X \sim \text{Poi}(\lambda)$ ,  $x \in \{0, 1, \dots\}$

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\begin{aligned} \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} &= e^{-\lambda} + e^{-\lambda} \cdot \lambda + e^{-\lambda} \cdot \frac{\lambda^2}{2!} + \dots \\ &= e^{-\lambda} \left( 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots \right) \\ &= e^{-\lambda} (e^{\lambda}) = 1 \end{aligned}$$

uniform distribution



$$F_x(a) = 0$$

$$F_x(b) = 1$$

$$f_x(x) = \frac{1}{b-a}$$

$$F_x(x) = (x-a) \times \frac{1}{b-a} = \frac{x-a}{b-a} \\ = P(X \leq x)$$

$$= \int_a^x f_x(x) dx$$

$$= \int_a^x \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \cdot [x]_a^x = \frac{x-a}{b-a}$$

$$E(x) = \int_a^b x \cdot f_x(x) dx$$

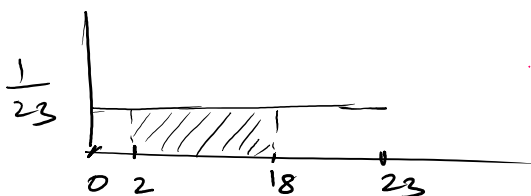
$$= \int_a^b x \left( \frac{1}{b-a} \right) dx$$

$$\Rightarrow V(x) = \int_a^b \left( x - \frac{a+b}{2} \right)^2 \cdot \frac{1}{b-a} dx$$

$$P(X=3)=0$$

$$a=0$$

$$b=23$$



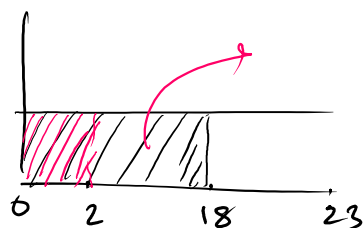
$$P(2 \leq X \leq 18)$$

$$= 16 \times \frac{1}{23} = \frac{16}{23}$$

$$F(x) = \frac{x-a}{b-a}$$

$$F(18) - F(2)$$

$$= \frac{18-0}{23-0} - \frac{2-0}{23-0} = \frac{16}{23}$$



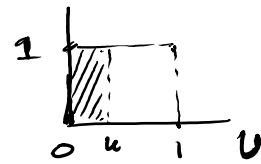
$$P(X > 12 | X > 8)$$

$$= \frac{P(X > 12, X > 8)}{P(X > 8)}$$

$$= \frac{P(X > 12)}{P(X > 8)} = \frac{1 - P(X < 12)}{1 - P(X < 8)}$$

$$= \frac{1 - \frac{12}{23}}{1 - \frac{8}{23}} = \frac{11}{15}$$

Q6.  $U \sim \text{unif}(0, 1)$



$$u \times 1 = u$$

$$\begin{aligned} \underline{\underline{F_u(u)}} &= \underline{\underline{u}}, & F_x(x) &= \frac{x-a}{b-a} \\ &= \frac{u-0}{1-0} \\ &= u \end{aligned}$$

$$X = aU + b,$$

$$F_x(x) = \mathbb{P}(X < x)$$

$$= \mathbb{P}(aU + b < x)$$

$$= \mathbb{P}\left(U < \frac{x-b}{a}\right)$$

$$= F_u\left(\frac{x-b}{a}\right) = \left(\frac{x-b}{a}\right) = \frac{\check{x-b}}{\underline{\underline{(a+b)-b}}}$$

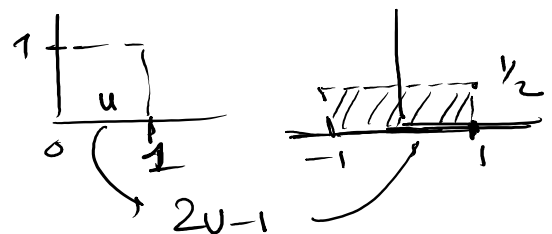
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$$\frac{x-\boxed{a}}{b-a}$$

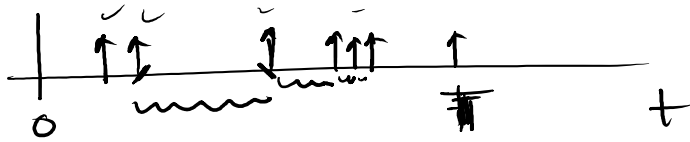
$$X \sim \text{Unif}(b, a+b)$$

$$X = 2U - 1, \quad a=2, \quad b=-1$$

$$X \sim \text{unif}(-1, 1)$$

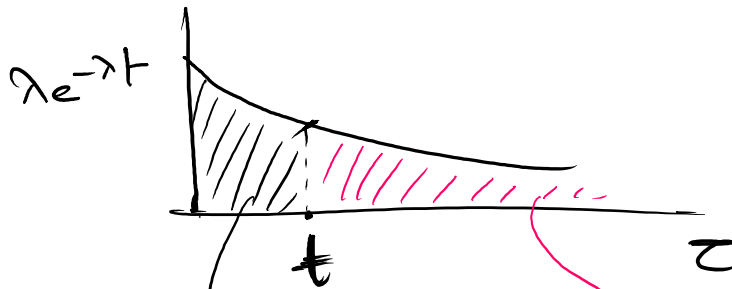


①



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Random Variable ( $\tau$ )  $\rightarrow$  time to default



$$\begin{aligned} P(\tau < t) &= F_{\tau}(t) \\ (\text{Cum PD}) \\ &= 1 - e^{-\lambda t} \end{aligned}$$

$$\begin{aligned} P(\tau > t) \\ (\text{Survival prob}) \\ &= e^{-\lambda t} \end{aligned}$$

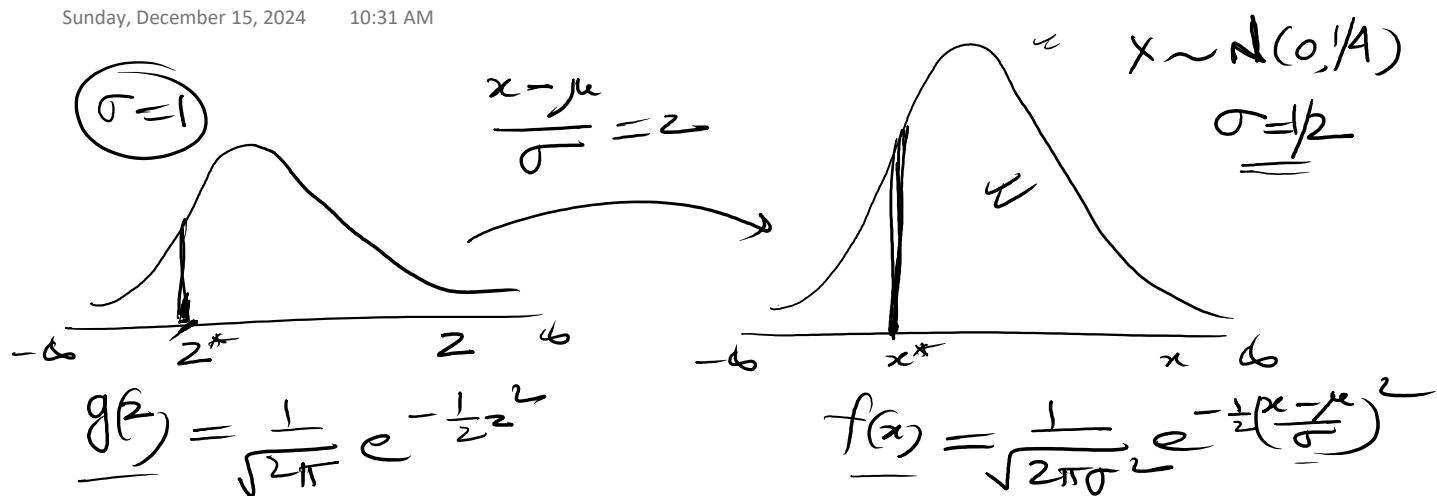
$$f(t) = \lambda e^{-\lambda t}$$

$$F(t) = \int_0^t f(s) ds$$

$$= \int_0^t \lambda e^{-\lambda s} ds$$

$$= \lambda \cdot \left[ \frac{e^{-\lambda s}}{-\lambda} \right]_0^t$$

$$= 1 - e^{-\lambda t}$$



$$\frac{g(z)}{\sigma} = f(x)$$

$$g(z) = \sigma f(x)$$

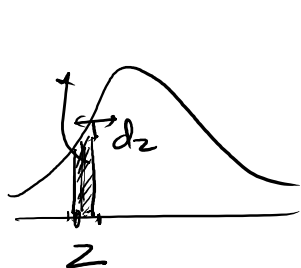
$$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}z^2}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

$$z = \mu + \frac{1}{2}z$$

$$z = -\frac{4}{2}, \left( x = -\frac{4}{2} \right)$$

$$z = 4, \left( x = \frac{4}{2} \right)$$



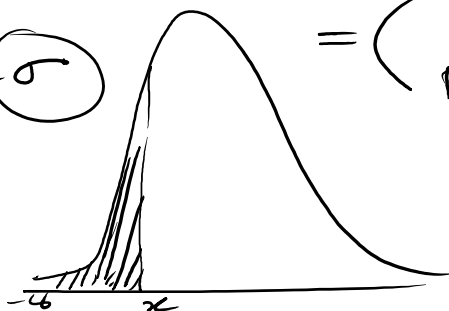
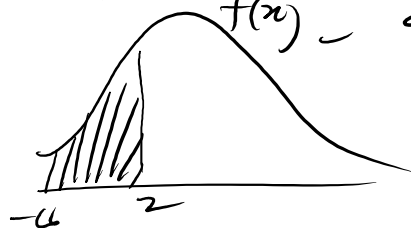
$$x = \mu + \sigma z$$

$$dx = \sigma dz$$

$$\left( \frac{dx}{dz} = \sigma \right)$$

$$g(z) dz = f(x) dx$$

$$\Rightarrow \frac{g(z)}{f(x)} = \frac{dx}{dz} = \sigma$$



$$= \begin{pmatrix} P(x < x) \\ P(Z < \frac{x-\mu}{\sigma}) \end{pmatrix}$$



$$Z_1 \sim N(0, 1)$$

$$Z_2 \sim N(0, 1)$$

$$\text{corr}(Z_1, Z_2) = \rho$$

$$\boxed{P(Z_1 > Z_2)} = P(\underline{Z_1 - Z_2} > 0) = P(\underline{Z^*} > 0) = \underline{0.5}$$

$$\parallel \underline{Z^*} \sim \underline{N(0, 1)}$$



$$f \begin{cases} X_1 \sim N(\mu_1, \sigma_1^2) \\ X_2 \sim N(\mu_2, \sigma_2^2) \end{cases}$$

$$\underline{aX_1 + bX_2} \sim \underline{N(\underline{a\mu_1 + b\mu_2}, \underline{a^2\sigma_1^2 + b^2\sigma_2^2 + 2ab\rho\sigma_1\sigma_2})}$$